Language of Instruction, Integration, and Identity: Evidence from Sweden's Tornedalians

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Another protest against Latvian-only education



Transition to Latvian-language instruction

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Ukraine



Russian-language instruction in occupied Ukraine, 2022–

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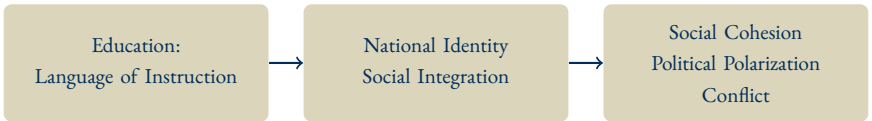
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This paper

- **Setting:** Introduction of Swedish-only schools in Finnish-speaking Tornedalen from 1889
- **Data:** Linked full-count individual-level censuses, 1880–1930

Two analyses:

1. **Individual exposure to Swedish-only schooling**
   - **Empirical strategy:** Variation in exposure across villages and cohorts
   - **Outcomes:** Social integration, national identity, and human capital
2. **Effects of school openings on the broader community**

Preview of Results (1/2)

Individual exposure to Swedish-only schooling:

- **Increased social integration**
  - Fewer Finnish-born partners, more partners from Tornedalen, and greater residential mixing
  - *Mechanism:* **intergroup contact** in schools
- **Shifted identity away from Swedish toward Finnish**
  - Higher Finnish and lower Swedish name scores, and lower military employment
  - *Mechanism:* language of instruction increasing the **salience of language**, changing perceived distance to Finns across the border, and altering social identity
- No effects on income, education beyond primary school, or language use at home

Preview of Results (2/2)

Effects of school openings on the broader community:

- **No evidence of broader cultural resistance**
  - Naming shifted in the opposite direction after school openings
  - No evidence of alternative schools opening
  - Finnish language use at home declined over time

Literature & Contribution

Nation-building through education (Blanc and Kubo, 2025; Hu and Luo, 2026; Clots-Figueras and Masella, 2013; Cantoni et al., 2017; Fouka, 2020; Lleras-Muney and Shertzer, 2015; Carlitz et al., 2025)

→ Individual schooling exposure vs. school openings on communities

Counter-assimilatory responses (Bisin and Verdier, 2001; Shayo, 2009, 2020; Meyersson, 2014; Atkin et al., 2021; Fouka and Tabellini, 2022, 2026; Fouka 2020; Carvalho et al., 2024; Maruthiah, 2025; Fouka and Serlin, 2026; Bazzi et al., 2026)

→ Social integration while weakening national identity

→ No evidence of broader cultural resistance

Language of instruction and education (Angrist and Lavy, 1997; Angrist et al., 2008; Taylor and von Fintel, 2016; Laitin and Ramachandran, 2016; Lloyd and Yang, 2025)

→ Limited human-capital effects

→ Role of language of instruction in shaping intergroup relations

Context and Data

Context

- Finland was part of Sweden for around 700 years
- The 1809 border with Russia split a culturally and linguistically connected region in two, largely along rivers
  - Cross-border interaction and movement nevertheless remained relatively easy
- The local language was historically classified as Finnish; today still spoken as Meänkieli



Tornedalen: Poor, Rural and Remote





Educational system **Before** 1889

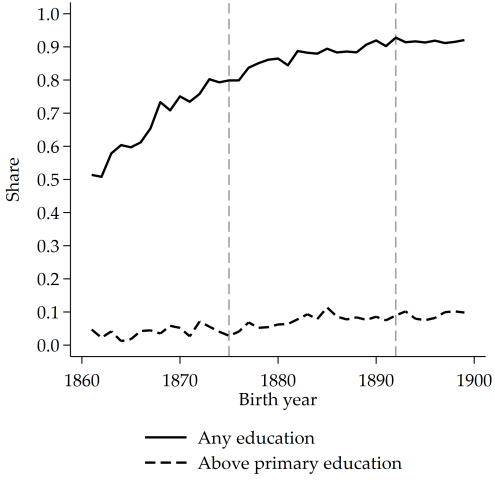
- Primary schooling mandated for **ages 7–14** but in reality low attendance
- Instruction was primarily in Finnish, while Swedish-language instruction was also provided in areas with Swedish-speaking children
- Subjects included arithmetic, **mother tongue (Swedish/Finnish)**, religion, history, geography, natural science, writing, drawing, singing, and gymnastics
- The traditional **"Translation Method"** used pupils' native language to teach Swedish, supported by textbooks with parallel Swedish–Finnish texts

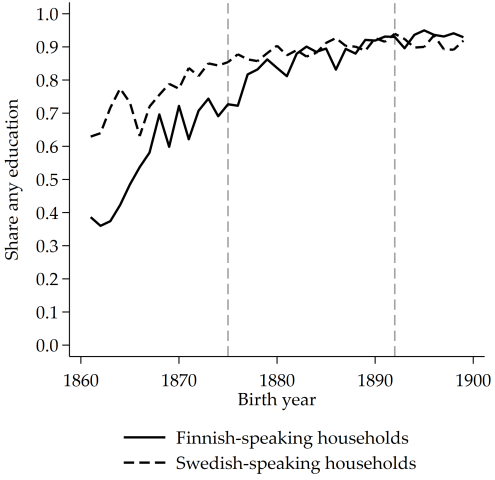
Educational System **After** 1889

- **State-funded primary schools** → Swedish-only schools
- Swedish as the sole language of instruction: **"All-Swedish Method"**
- Testimonies of punishment and shame (Truth and Reconciliation Commission, 2023)
- Parish-level participation was **voluntary** and motivated by the **economic benefits**



Educational Attainment





Source: Swedish population censuses for Tornedalen.

Data: Full-count Swedish censuses, 1880–1930

- **Link individuals over time**
  - Pre-treatment characteristics in 1890; follow individuals to 1910 and 1930
- **Key information**
  - Village of residence, geocoded by hand
  - Pre-treatment education supply from teacher occupations
  - Recorded as "Finnish," largely based on language use at home, 1880–1910
  - Education (self-reported) and income (tax registers) 1930
  - Family links to partners and children

Linking methodology → Linkage rates → Is the linked sample representative? →

Measuring Social Integration

1. **Marriage patterns:**
   - Partner born in Finland
   - Partner born in Tornedalen
   - Partner born in Sweden, but not Tornedalen (Swe. excl. Torn.)
2. **Village-level ethnolinguistic fractionalization**
   - The index measures how likely two randomly chosen individuals are to belong to different ethnolinguistic groups (Alesina et al., 2003).

$$F_l = 1 - (p_l^2 + (1 - p_l)^2)$$

   where p_l is the share "Finnish" in village l (recorded as "Finnish" or born in Finland)
   - *Interpretation:* $F_l = 0$ if the village is fully homogeneous, maximized at $p_l = 0.5$

Measuring National Identity

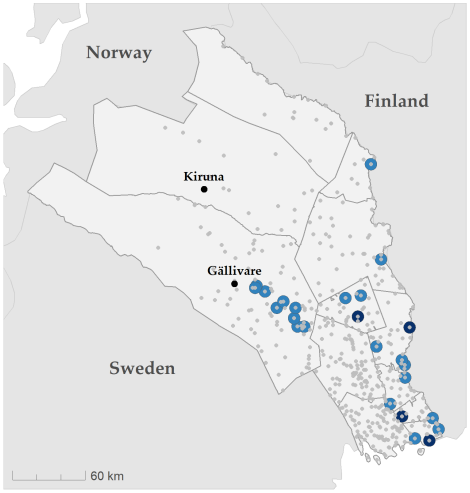
1. **Children's names**
   - Proxy for cultural transmission; used to measure individualism (Knudsen, 2019; Bazzi et al., 2020), religiosity (Andersen & Bentzen, 2025), and ethnic identity (Fryer & Levitt, 2004; Bazzi et al., 2019)
   - Cultural Name Score (CNS) for each name n, cultural group g, and year t.
   - Let G denote the five cultural groups: Swedish, Finnish, Tornedalian, Sámi, and Immigrant.

$$CNS_{ngt} = 100 \times \frac{P_t(n \mid g)}{\sum_{h \in G} P_t(n \mid h)}$$

2. **Working in the military**
   - High-stakes commitment to the nation
   - Military participation is a common proxy for national attachment (Fouka, 2020; Blanc & Kubo, 2025; Qian & Tabellini, 2026)

CNS distributions 1890 → | Examples of high scoring names →

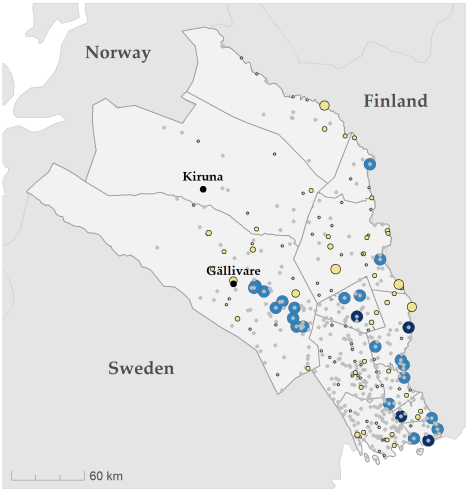
Villages in 1890, Swedish-Only Schools & Treatment Assignment



	Not treated	Matched control	Treated
Economic and demographic			
Population (mean)	76.07	192.26	175.66
Education			
Any teacher 1890	0.11	0.30	0.28
Culture			
Finnish at home	0.46	0.73	0.72
Finnish CNS (mean)	18.26	19.29	19.70
Swedish CNS (mean)	18.66	17.89	17.86
Distance to Finnish border (log)	3.50	2.79	2.62
Villages	405	104	47

Full table →

Villages in 1890, Swedish-Only Schools & Treatment Assignment



Swedish-only schools (5 km radius) 1889 1895

Matching weight Villages 1890 1 2 3

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Full table →

Exposure to Swedish-only Schooling

Identification

- **Goal:** Estimate causal effect of exposure to Swedish-only schooling
- **Bundled treatment:**
  - Language of instruction
  - Access to schooling
- **Two sources of variation:**
  - Comparing villages with and without a Swedish-only school
  - Comparing exposed cohorts (≤ 14) to not-exposed cohorts (> 14)
- **Assumption**: in the absence of Swedish-only schools, cohort differences would have evolved in parallel in treated and control villages

Empirical Specification

$$y_{i,l,c} = \beta \left(\text{treated_village}_{l} \times \text{school_age}_{l,c} \right) + \alpha_l + \alpha_c + \epsilon_{i,l,c}.$$

- $y_{i,l,c}$: outcome for individual i from village l, birth cohort c
  1. **social integration**
  2. **national identity**
  3. education, income and language use at home
- $\text{treated_village}_{l} = 1$ if village l is within 5 km of a Swedish-only school.
- $\text{school_age}_{l,c} = 1$ if cohort c was age 3–14 when the school opened (control ages: 15–22).
- Staggered introduction (1889 and 1895) → **stacked diff-in-diff** (Wing et al., 2024).

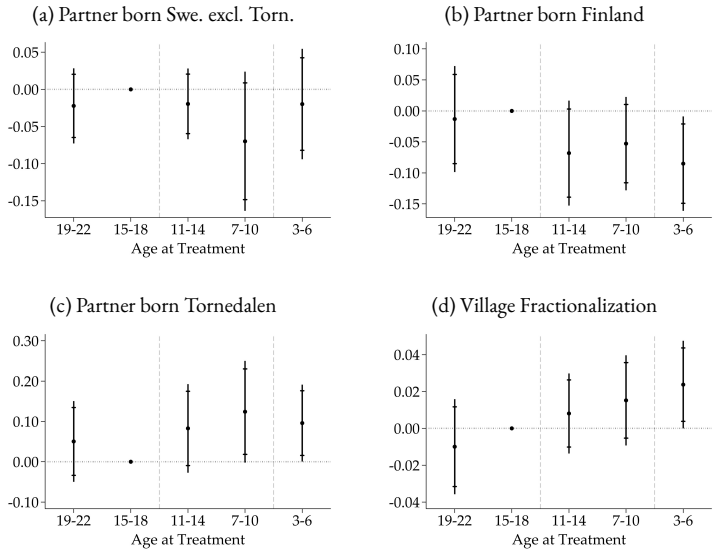
Placebo outcomes in 1890 →

1. Swedish-only Schooling and Social Integration

	Partner born Swe. excl. Torn. (1)	Partner born Tornedalen (2)	Partner born Finland (3)	Village fractionalization (4)
Panel A.				
Treated	-0.026	0.077**	-0.063**	0.021**
	(0.027)	(0.037)	(0.025)	(0.010)
Panel B.				
Years treated	-0.003	0.008**	-0.007***	0.002**
	(0.004)	(0.004)	(0.003)	(0.001)
Village-Stack FE	✓	✓	✓	✓
Age-Stack FE	✓	✓	✓	✓
Dep. var. mean	0.077	0.846	0.090	0.128
N	4,069	4,069	4,069	5,700

Village-level minority share → Migration → Intermarriage based on predicted ethnicity →

1. Swedish-only Schooling and Social Integration



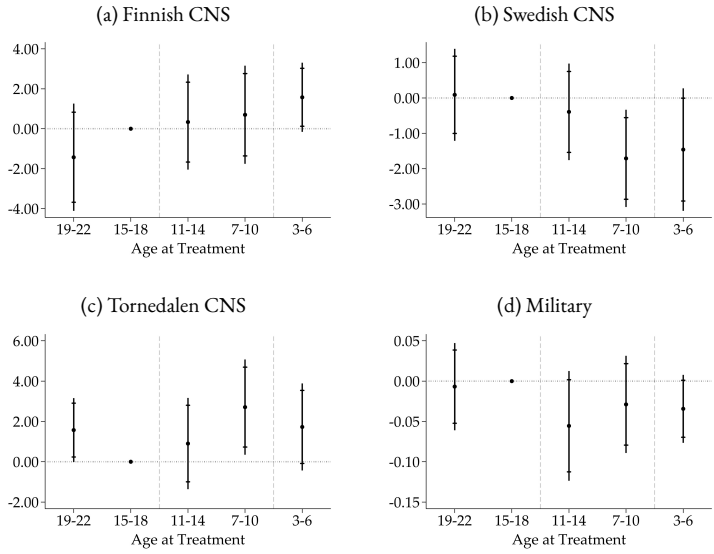
2. Swedish-only Schooling and National Identity

	Finnish CNS	Swedish CNS	Tornedalian CNS	Military
	(1)	(2)	(3)	(4)
Panel A.				
Treated	1.539**	-1.237**	1.057	-0.035**
	(0.707)	(0.509)	(0.774)	(0.014)
Panel B.				
Years treated	0.224**	-0.154**	0.120	-0.001
	(0.092)	(0.075)	(0.078)	(0.001)
Village-Stack FE	✓	✓	✓	✓
Age-Stack FE	✓	✓	✓	✓
Dep. var. mean	20.520	20.458	26.704	0.029
N	3,925	3,925	3,925	3,071

Effect sizes: 6% decrease in Swedish CNS and a 7.5% increase in Finnish CNS relative to their unconditional means.

Alternative Majority CNS →

2. Swedish-only Schooling and National Identity



3. Swedish-only Schooling, Education, Income and Language Use at Home

	Any education (1)	Further education (2)	Any income (3)	Log income (4)	Finnish at home (5)
Panel A.					
Treated	0.041	-0.009	-0.026	-0.007	-0.014
	(0.048)	(0.024)	(0.037)	(0.096)	(0.034)
Panel B.					
Years treated	-0.001	-0.001	-0.001	0.003	-0.000
	(0.004)	(0.002)	(0.003)	(0.014)	(0.003)
Panel C.					
Treated	0.140***	0.004	-0.024	0.034	-0.011
	(0.053)	(0.022)	(0.046)	(0.150)	(0.047)
Treated × Teacher before	-0.162***	-0.020	-0.005	-0.063	-0.005
	(0.040)	(0.040)	(0.062)	(0.158)	(0.038)
Village-Stack FE	✓	✓	✓	✓	✓
Age-Stack FE	✓	✓	✓	✓	✓
Dep. var. mean	0.869	0.081	0.438	7.241	0.443
N	4,163	4,163	4,163	1,917	5,133

Results men → Results women → Occupational upgrading →

Threats to Identification

- **Composition**
  - No differential *emigration* →
  - No differential propensity to have a *partner* →
  - No differential propensity to have a *child* →
- **Concurrent events**
  - Robust to differential effects by *distance to Gällivare* →
  - Robust to differential effects by *distance to the Finnish border* →
- **Spatial correlation**
  - Robust to *Conley SEs* (3, 5, and 10 km) →

Mechanisms

Taking Stock and What Comes Next

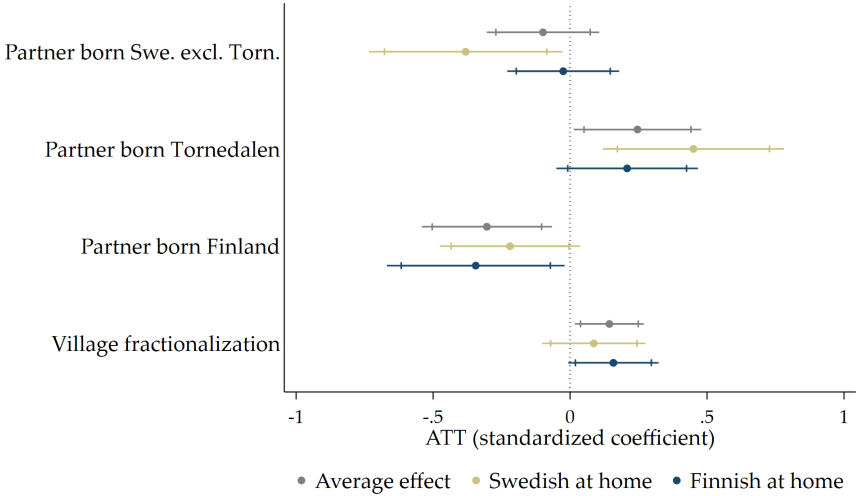
- **Taking stock:** Swedish-only schooling increased social integration within the region while shifting national identity away from Swedish toward Finnish.

- I now examine mechanisms behind:
  1. **social integration**
  2. **national identity**

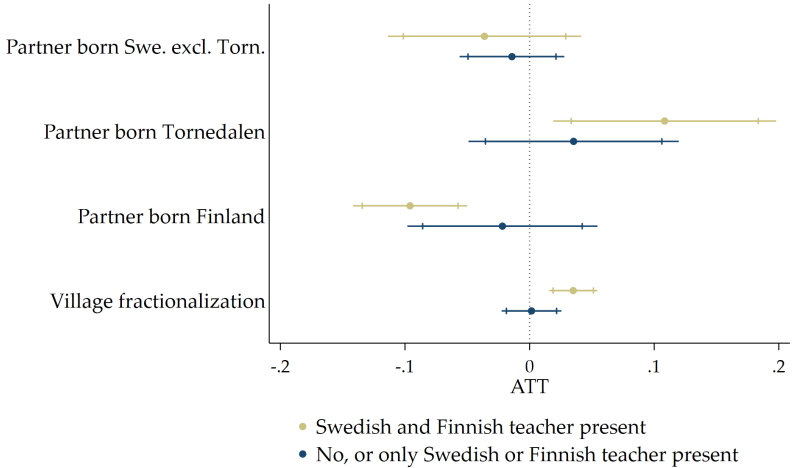
Mechanisms: Social Integration

- **Suggested mechanisms:** language skills vs. intergroup contact
  - Null effects on human capital and language use at home make language skills *alone* less likely
- **Theory:** intergroup contact may increase integration through networks and reduced prejudice (Allport, 1954; Lowe, 2025).
- **Testable predictions:**
  - *Language skills*
    - → Larger effects for Finnish-speaking households
  - *Intergroup contact*
    - → Effects for both Finnish- and Swedish-speaking households
    - → Larger effects where pre-reform language separation was greater

Social Integration - Heterogeneity by Language Use at Home



Social Integration - Heterogeneity by Teachers Present Within 10 km



Mechanism: National Identity

- **Suggested mechanism:** social identity

  Swedish-only instruction → language becomes more salient as an ethnic attribute → perceived distance to Swedes ↓ and Finns ↑

- **Theory:** social identity depends on the social context (Shayo, 2009, 2020).
  1. *Group status* (Atkin et al., 2021)
  2. *Perceived distance* to other group members, which may depend on the salience of group attributes (Fouka and Tabellini, 2026)

- **Testable prediction:**
  - Stronger Finnish naming among individuals with less prior exposure to Swedish instruction

Heterogeneity by recorded language use at home →

National Identity - Distance to a Swedish Teacher

	Finnish CNS		
	(1)	(2)	(3)
Treated	0.084	0.097	0.940
	(0.781)	(0.830)	(1.616)
Treated × Distance to Swedish teacher	0.765***	0.777**	0.775**
	(0.256)	(0.308)	(0.309)
Treated × Distance to Finnish teacher		-0.028	-0.084
		(0.386)	(0.372)
Treated × Share Finnish at home			-1.001
			(1.778)
Village-Stack FE	✓	✓	✓
Age-Stack FE	✓	✓	✓
Dep. var. mean	20.520	20.520	20.520
N	3,925	3,925	3,925

Cultural Resistance as an Alternative Mechanism

- **Alternative mechanism:** cultural resistance (Bisin and Verdier, 2001; Fouka, 2020; Carvalho et al., 2024).
- **Theory:** parents and communities transmit identity and culture across generations
  - Transmission may strengthen in response to assimilation pressure
- **Testable predictions:**
  - Larger effects where the minority group is smaller →
  - Community-level responses consistent with cultural resistance:
    1. Decreased Swedish naming after school openings
    2. Decreased Swedish-language use in villages
    3. Opening of alternative schools (Bazzi et al., 2026)

Effects of school openings on the broader community

Empirical Specification: Children's CNSs Around School Opening

$$\text{CNS}_{i,l,c} = \sum_{j \notin [1,2]} \beta_j \left(\text{treated_village}_l \times \text{age_at_opening}_{j(l,c)} \right) + \alpha_c + \alpha_l + \alpha_f + \epsilon_{i,l,c}$$

- $\text{CNS}_{i,l,c}$: Swedish or Finnish name score for child i
- $\text{treated_village}_l = 1$ if village l is within 5 km of a Swedish-only school
- $\text{age_at_opening}_{j(l,c)}$: two-year age-at-opening bins
- α_f: family fixed effects

Children's CNSs Around School Opening

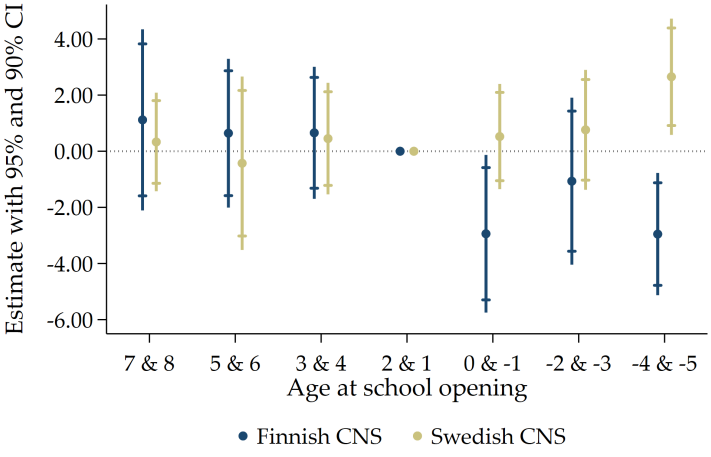


Table →

Empirical Specification: Village-Level Effects

$$y_{l,t} = \beta \left(\text{treated_village}_l \times \text{post}_t \right) + \alpha_l + \alpha_t + \varepsilon_{l,t}$$

- $y_{l,t}$: outcome for village l in census year t
  1. Presence of Swedish and Finnish teachers within 5 km
  2. Share speaking Finnish at home
- $\text{treated_village}_l = 1$ if village l is within 5 km of a Swedish-only school
- $\text{post}_t = 1$ in 1900 and 1910; 1880 and 1890 are the pre-treatment years

Village-Level Effects

	Any Swedish teacher	Any Finnish teacher	Share Finnish at home
	(1)	(2)	(3)
Treated × Post	0.285***	-0.004	-0.102**
	(0.071)	(0.074)	(0.048)
Dep. var. mean	0.344	0.281	0.677
Observations	604	604	577
Villages	151	151	150
Village FE	✓	✓	✓
Year FE	✓	✓	✓

Event studies → Other outcomes → Birthplace shares →

Conclusion

Takeaways:

- Exposure to Swedish-only schooling increased social integration within the region, but shifted expressions of identity from Swedish toward Finnish.
- No evidence of broader cultural resistance to the schools.
- The identity response to schooling did not prevent the decline in Finnish language use at home in villages over time.

Policy implications:

- Joint schooling may matter more for integration than imposing a particular language of instruction.
- Language-of-instruction reforms aimed at improving learning should also consider their effects on intergroup contact.

Thank you!

Linking the Census Over Time

Automated record linkage (Abramitzky et al., 2021; Berger, Karadja & Prawitz, 2024)

- Exact match on **year of birth, parish of birth, and sex**
- Unique match within Jaro–Winkler distance < 0.15 (< 0.10) for **first and last names**

Main problem: children do not have a recorded surname

- Impute the **household head's surname**
- Construct a **patronymic surname** (father's name + "dotter" or "son")

Back →

Linkage Rates

	Male	Female	All
1910 full sample	81.3	81.8	81.5
1910 sample directly linked	82.5	82.6	82.6
1910 sample linked through parents	71.6	75.1	73.3
1930 full sample	82.4	82.1	82.2
1930 sample directly linked	83.4	83.5	83.4
1930 sample linked through parents	73.3	70.0	71.7

Back →

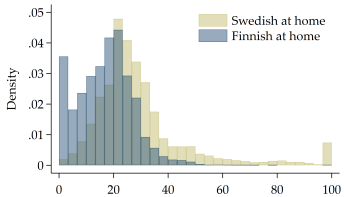
Is the Sample Representative?

	Full 1890	Linked to 1910	Linked to 1930
Female	0.49	0.48	0.48
Finnish at home	0.49	0.48	0.47
Servants in household	0.13	0.14	0.13
Household head in labor force	0.80	0.81	0.81
Farm household	0.57	0.58	0.60
Household size	6.58	6.76	6.78
Father in household	0.91	0.93	0.93
N	22,195	13,667	10,783

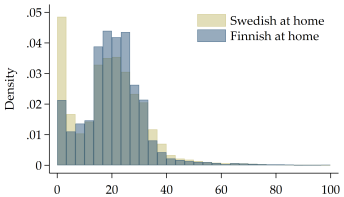
Back →

Distribution of CNS 1890

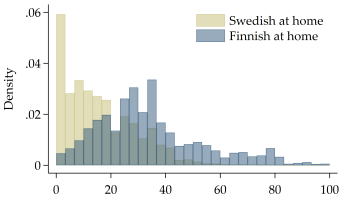
(a) Swedish CNS



(b) Finnish CNS



(c) Tornedalian CNS



Back →

Examples of Names with High Cultural Name Scores

Swedish

Male		Female	
Name	CNS	Name	CNS
ARVID	100	LINNEA	100
EINAR	100	SIGNE	100
RAGNAR	100	ASTRID	100
GOSTA	100	RUT	100
BERTIL	100	EDIT	100
ALGOT	100	GERDA	100
FOLKE	100	SVEA	100
BIRGER	100	VIOLA	100
HILDING	100	MARGIT	100
TURE	100	ALFRIDA	100

Finnish

Male		Female	
Name	CNS	Name	CNS
MATTI	100	SUOMA	100
ARMAS	100	SAIDA	100
JUHA	100	LYGLI	100
PEKKA	100	SAIMI	100
ANTTI	100	DORDI	100
JUHO	100	KATRI	100
TOIVO	100	AINO	100
VAINO	100	XENIA	100
ARVO	100	IRJA	100
KAARLE	100	ADELAID	100

Tornedalian

Male		Female	
Name	CNS	Name	CNS
VALDE	100	MEIMI	100
ILMAR	100	SENEFIA	100
HEIMAR	100	VESTINA	100
FALE	99	NETTI	100
KALEB	99	HELLI	100
DONATUS	99	GENSINA	100
OLGE	99	TYYNI	99
ABNER	99	NETTY	99
AMBERT	99	AINI	99
HYGIENUS	99	MILIANA	99

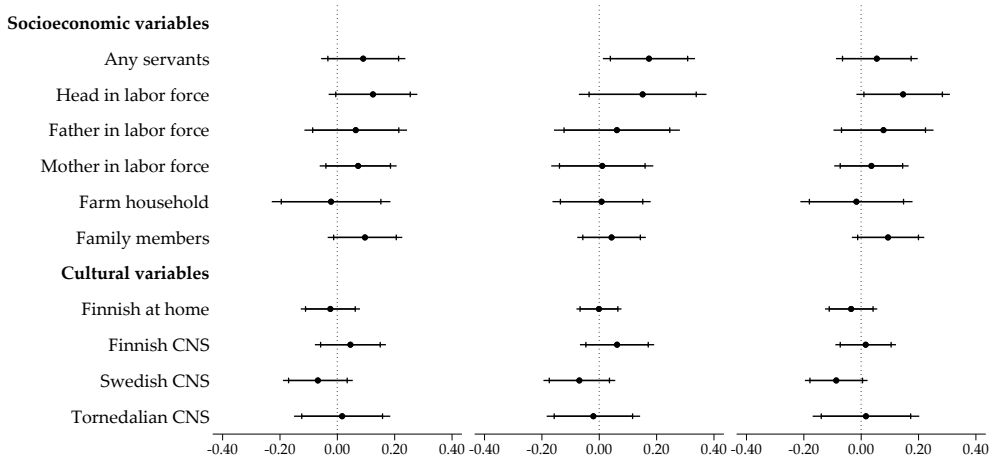
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Villages in 1890: Descriptive Statistics

	Not treated	Matched control	Treated
Economic and demographic			
Population (mean)	76.07	192.26	175.66
Population (log)	3.48	4.55	4.42
Age (mean)	25.91	25.21	26.22
Males (share)	0.52	0.51	0.51
Females in the labor force (share)	0.08	0.06	0.07
Males in the labor force (share)	0.49	0.48	0.54
Farm households (share)	0.65	0.53	0.58
Household members (mean)	6.16	5.61	5.55
Distance to Gällivare (log)	4.71	4.52	4.72
Education			
Any teacher 1880	0.07	0.09	0.13
Any teacher 1890	0.11	0.30	0.28
Culture			
Finnish at home	0.46	0.73	0.72
Finnish CNS (mean)	18.26	19.29	19.70
Swedish CNS (mean)	18.66	17.89	17.86
Tornedalian CNS (mean)	28.12	32.02	32.39
Distance to Finnish border (log)	3.50	2.79	2.62
Villages	405	104	47

Back →

Placebo: Outcomes Measured in 1890



Back →

Minority Share in Village, 1910

	Share Finnish in village	
	(1)	(2)
Treated	0.016	
	(0.031)	
Years treated		0.002
		(0.003)
Village-Stack FE	✓	✓
Age-Stack FE	✓	✓
Dep. var. mean	0.456	0.456
N	5,700	5,700

Back →

Swedish-only Schools and Migration

	Stayed in village	Mig. dist. (km)	ln(1 + mig. dist.)	Left Tornedalen
	(1)	(2)	(3)	(4)
Panel A.				
Treated	0.084**	-6.945**	-0.288**	-0.004
	(0.037)	(3.280)	(0.145)	(0.021)
Panel B.				
Years treated	0.006	-0.480	-0.021	-0.002
	(0.004)	(0.409)	(0.016)	(0.002)
Village-Stack FE	✓	✓	✓	✓
Age-Stack FE	✓	✓	✓	✓
Dep. var. mean	0.468	29.413	1.849	0.113
N	6,168	6,168	6,168	6,456

Back →

Swedish-Only Schooling and Partner (Predicted) Ethnicity

	Predicted Finnish attachment		Predicted ethnic mixedness	
	(1)	(2)	(3)	(4)
Treated	-0.035		0.035*	
	(0.040)		(0.021)	
Years treated		-0.004		0.004**
		(0.004)		(0.002)
Village-Stack FE	✓	✓	✓	✓
Age-Stack FE	✓	✓	✓	✓
Dep. var. mean	0.620	0.620	0.185	0.185
N	3,619	3,619	3,619	3,619

Notes: Partner ethnicity is predicted using the ethnic composition associated with the partner's surname in the 1890 census.

Back →

Swedish-Only Schooling and Majority CNS

	Majority CNS	
	(1)	(2)
Treated	-2.128**	
	(0.879)	
Years treated		-0.264**
		(0.113)
Village-Stack FE	✓	✓
Age-Stack FE	✓	✓
Dep. var. mean	46.298	46.298
N	3,925	3,925

Notes: Majority CNS measures how distinctively Swedish a name is relative to non-Swedish names. All alternative cultural groups are combined into a single reference category.

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Education, Income and Language Use at Home: Men

	Any education (1)	Further education (2)	Any income (3)	Log income (4)	Finnish at home (5)
Panel A.					
Treated	0.056	-0.007	-0.008	-0.106	0.001
	(0.055)	(0.030)	(0.047)	(0.106)	(0.040)
Panel B.					
Years treated	0.003	-0.000	0.001	-0.008	-0.001
	(0.004)	(0.003)	(0.005)	(0.015)	(0.003)
Panel C.					
Treated	0.188***	0.006	0.038	-0.049	0.008
	(0.067)	(0.027)	(0.060)	(0.155)	(0.054)
Treated × Teacher before	-0.220***	-0.022	-0.078	-0.091	-0.010
	(0.064)	(0.041)	(0.075)	(0.166)	(0.055)
Village-Stack FE	✓	✓	✓	✓	✓
Age-Stack FE	✓	✓	✓	✓	✓
Dep. var. mean	0.866	0.082	0.749	7.247	0.447
N	2,152	2,152	2,152	1,665	2,706

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Education, Income and Language Use at Home: Women

	Any education (1)	Further education (2)	Any income (3)	Log income (4)	Finnish at home (5)
Panel A.					
Treated	0.031	-0.011	0.002	-0.099	-0.017
	(0.050)	(0.040)	(0.029)	(0.295)	(0.049)
Panel B.					
Years treated	-0.005	-0.003	0.003	0.015	0.002
	(0.004)	(0.004)	(0.003)	(0.032)	(0.005)
Panel C.					
Treated	0.092	-0.018	0.012	-1.258**	-0.009
	(0.074)	(0.029)	(0.035)	(0.540)	(0.063)
Treated × Teacher before	-0.097	0.011	-0.016	1.277**	-0.013
	(0.069)	(0.052)	(0.053)	(0.526)	(0.069)
Village-Stack FE	✓	✓	✓	✓	✓
Age-Stack FE	✓	✓	✓	✓	✓
Dep. var. mean	0.873	0.080	0.108	7.242	0.439
N	1,993	1,993	1,993	221	2,413

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Swedish-only Schooling and Occupational Upgrading

	Any job	Farmer or fisher	Skill level	Higher skilled	Medium skilled	Lower skilled	Unskilled
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A. Pooled							
Treated	0.012	0.029	0.094	0.017	0.015	-0.044	-0.017
	(0.026)	(0.044)	(0.127)	(0.040)	(0.038)	(0.035)	(0.034)
Dep. var. mean	0.579	0.258	2.235	0.119	0.183	0.192	0.248
N	5,723	3,328	2,396	3,328	3,328	3,328	3,328
Panel B. Men							
Treated	0.003	0.035	0.007	0.013	-0.037	-0.006	-0.005
	(0.035)	(0.049)	(0.146)	(0.039)	(0.038)	(0.026)	(0.038)
Dep. var. mean	0.828	0.316	2.290	0.140	0.145	0.172	0.227
N	3,012	2,597	1,739	2,597	2,597	2,597	2,597
Panel C. Women							
Treated	0.034	0.001	0.305	0.081	0.129	-0.245**	0.034
	(0.038)	(0.058)	(0.253)	(0.084)	(0.084)	(0.094)	(0.081)
Dep. var. mean	0.307	0.083	2.117	0.060	0.295	0.252	0.309
N	2,697	710	636	710	710	710	710

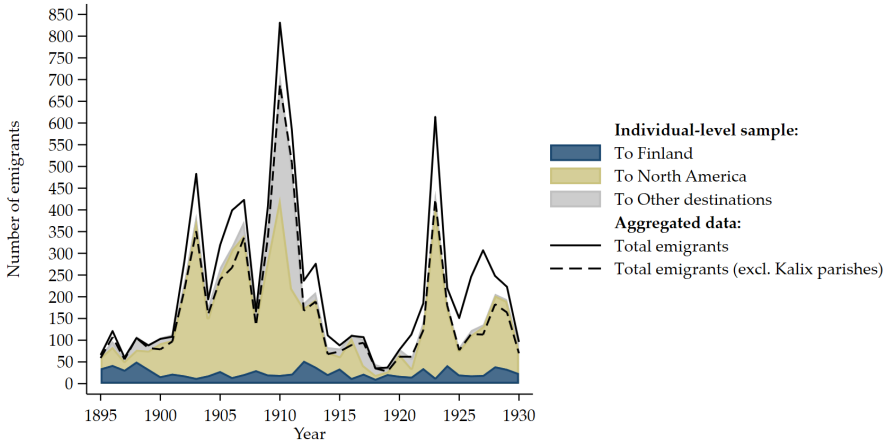
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Emigration

	Any country		Finland	
	(1)	(2)	(3)	(4)
Treated	-0.019		-0.001	
	(0.041)		(0.014)	
Years treated		-0.001		0.002
		(0.004)		(0.002)
Village-Stack FE	✓	✓	✓	✓
Age-Stack FE	✓	✓	✓	✓
Dep. var. mean	0.126	0.126	0.020	0.020
N	5,194	5,194	5,194	5,194

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Emigration over Time



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Conley Standard Errors

	Children's CNS		Works in the	Partner born in		Village level
	Finnish	Swedish	military	Tornedalen	Finland	fractionalization
Panel A: Conley bandwidth = 5 km						
Treated	1.539**	-1.237**	-0.035**	0.077**	-0.063***	0.021**
	(0.710)	(0.500)	(0.014)	(0.037)	(0.024)	(0.010)
Panel B: Conley bandwidth = 3 km						
Treated	1.539**	-1.237**	-0.035**	0.077**	-0.063**	0.021**
	(0.702)	(0.505)	(0.014)	(0.038)	(0.025)	(0.009)
Panel C: Conley bandwidth = 10 km						
Treated	1.539**	-1.237**	-0.035***	0.077**	-0.063***	0.021**
	(0.719)	(0.500)	(0.013)	(0.036)	(0.024)	(0.010)
Dep. var. mean	20.516	20.459	0.029	0.846	0.090	0.128
Village-Stack FE	✓	✓	✓	✓	✓	✓
Age-Stack FE	✓	✓	✓	✓	✓	✓

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Concurrent Event: Gällivare

	Finnish CNS		Swedish CNS		Tornedalian CNS		Military	
Panel A: National Identity								
Treated		1.497**		-1.394***		1.292		-0.033**
		(0.743)		(0.490)		(0.818)		(0.014)
Age ≤ 14 × Distance to Gällivare	0.163	0.067	0.165*	0.254***	-0.296	-0.379*	-0.005**	-0.003
	(0.200)	(0.232)	(0.098)	(0.074)	(0.196)	(0.195)	(0.002)	(0.002)
Dep. var. mean	20.520	20.520	20.458	20.458	26.704	26.704	0.029	0.029
N	3,925	3,925	3,925	3,925	3,925	3,925	3,071	3,071
	Partner born Finland		Partner born Tornedalen		Partner born Swe. excl. Torn.		Village fractionalization	
Panel B: Social integration								
Treated		-0.061**		0.069*		-0.018		0.021**
		(0.027)		(0.037)		(0.025)		(0.010)
Age ≤ 14 × Distance to Gällivare	-0.006	-0.002	0.018***	0.014**	-0.014***	-0.013***	0.002	0.001
	(0.005)	(0.005)	(0.006)	(0.006)	(0.004)	(0.003)	(0.003)	(0.003)
Dep. var. mean	0.090	0.090	0.846	0.846	0.077	0.077	0.128	0.128
N	4,069	4,069	4,069	4,069	4,069	4,069	5,700	5,700
Village-Stack FE	✓	✓	✓	✓	✓	✓	✓	✓
Age-Stack FE	✓	✓	✓	✓	✓	✓	✓	✓

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Concurrent Event: Border

	Finnish CNS		Swedish CNS		Tornedalian CNS		Military	
Panel A: National identity								
Treated		2.128***		-1.552***		1.469		-0.033**
		(0.715)		(0.480)		(0.929)		(0.013)
Age ≤ 14 × Distance to border	0.592*	0.803***	-0.276	-0.430***	0.415	0.561	0.005	0.003
	(0.353)	(0.307)	(0.212)	(0.147)	(0.390)	(0.403)	(0.006)	(0.004)
Dep. var. mean	20.520	20.520	20.458	20.458	26.704	26.704	0.029	0.029
N	3,925	3,925	3,925	3,925	3,925	3,925	3,071	3,071
	Partner born Finland		Partner born Tornedalen		Partner born Swe. excl. Torn.		Village fractionalization	
Panel B: Social integration								
Treated		-0.063*		0.070*		-0.016		0.014*
		(0.032)		(0.039)		(0.023)		(0.008)
Age ≤ 14 × Distance to border	0.006	0.000	-0.017	-0.011	0.016*	0.014*	-0.010***	-0.009***
	(0.013)	(0.015)	(0.016)	(0.016)	(0.009)	(0.007)	(0.003)	(0.003)
Dep. var. mean	0.090	0.090	0.846	0.846	0.077	0.077	0.128	0.128
N	4,069	4,069	4,069	4,069	4,069	4,069	5,700	5,700
Village-Stack FE	✓	✓	✓	✓	✓	✓	✓	✓
Age-Stack FE	✓	✓	✓	✓	✓	✓	✓	✓

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Outcome: Has a Partner

	Female & Male		Female		Male	
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	-0.029		-0.015		-0.038	
	(0.035)		(0.056)		(0.051)	
Years treated		-0.005		-0.002		-0.007
		(0.005)		(0.007)		(0.006)
Village-Stack FE	✓	✓	✓	✓	✓	✓
Age-Stack FE	✓	✓	✓	✓	✓	✓
Dep. var. mean	0.638	0.638	0.648	0.648	0.628	0.628
N	6,456	6,456	3,025	3,025	3,415	3,415

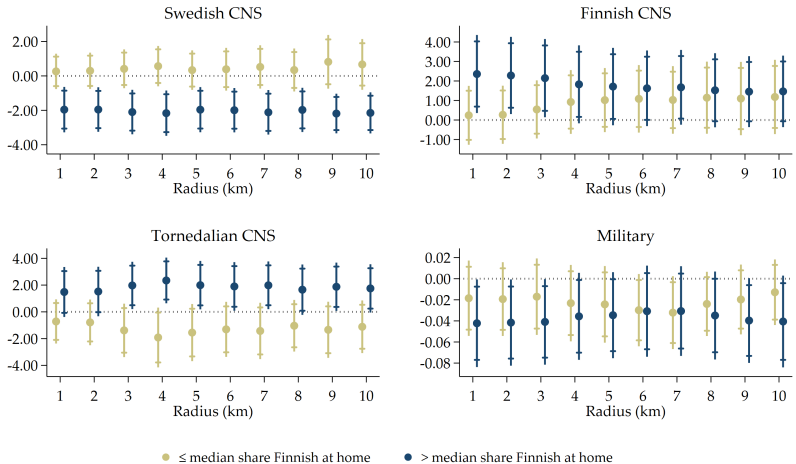
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Outcome: Has a Child

	Female & Male		Female		Male	
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	-0.049		-0.056		-0.045	
	(0.039)		(0.053)		(0.054)	
Years treated		-0.005		-0.004		-0.007
		(0.005)		(0.007)		(0.006)
Village-Stack FE	✓	✓	✓	✓	✓	✓
Age-Stack FE	✓	✓	✓	✓	✓	✓
Dep. var. mean	0.623	0.623	0.664	0.664	0.586	0.586
N	6,456	6,456	3,025	3,025	3,415	3,415

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National Identity – Heterogeneity by Share Recorded as "Finnish"



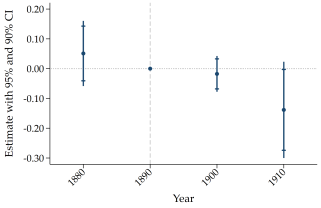
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CNSs Around School Opening

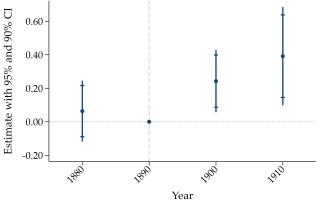
	Female & Male			Female	Male
	(1)	(2)	(3)	(4)	(5)
Panel A: Finnish CNS					
Years since opening	-0.216**	-0.298**	-0.459***	-0.196	-0.278*
	(0.109)	(0.126)	(0.156)	(0.186)	(0.141)
Dep. var. mean	20.398	20.321	20.321	20.859	19.946
N	11,378	9,438	9,438	5,554	5,803
Panel B: Swedish CNS					
Years since opening	0.270***	0.304***	0.338***	0.178	0.355***
	(0.082)	(0.079)	(0.128)	(0.152)	(0.115)
Dep. var. mean	21.511	21.464	21.464	22.025	21.018
N	11,378	9,438	9,438	5,554	5,803
Panel C: Tornedalian CNS					
Years since opening	-0.095	-0.109	-0.332*	0.052	-0.270
	(0.134)	(0.132)	(0.172)	(0.220)	(0.213)
Dep. var. mean	26.021	26.183	26.183	25.076	26.935
N	11,378	9,438	9,438	5,554	5,803
Village-Stack FE	✓	✓	✓	✓	✓
Birthyear-Stack FE	✓	✓	✓	✓	✓
Sibling sample		✓	✓		
Sibling FE			✓		

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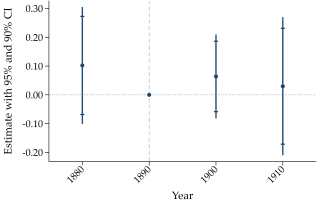
Village-Level Effects: Event Studies



Share Finnish at home



Any Swedish teacher



Any Finnish teacher

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Village-Level Social Integration and Identity

Panel A. Social integration

	Village fractionalization	Fin. – Swe. (incl. Torn.) couple	Swe. (excl. Torn.) – Fin./Torn. couple	Couple born in Torn.
	(1)	(2)	(3)	(4)
Treated × Post	-0.010	0.021	0.016	-0.031
	(0.024)	(0.026)	(0.013)	(0.040)
Dep. var. mean	0.096	0.103	0.032	0.831
Observations	577	564	564	564
Villages	150	149	149	149

Panel B. National dentity

	Military	Tornedalian CNS	Finnish CNS	Swedish CNS
	(1)	(2)	(3)	(4)
Treated × Post	0.007***	0.741	-0.548	0.333
	(0.003)	(0.922)	(0.616)	(0.571)
Dep. var. mean	0.004	30.581	18.922	18.179
Observations	576	560	560	560
Villages	150	149	149	149
Village FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓

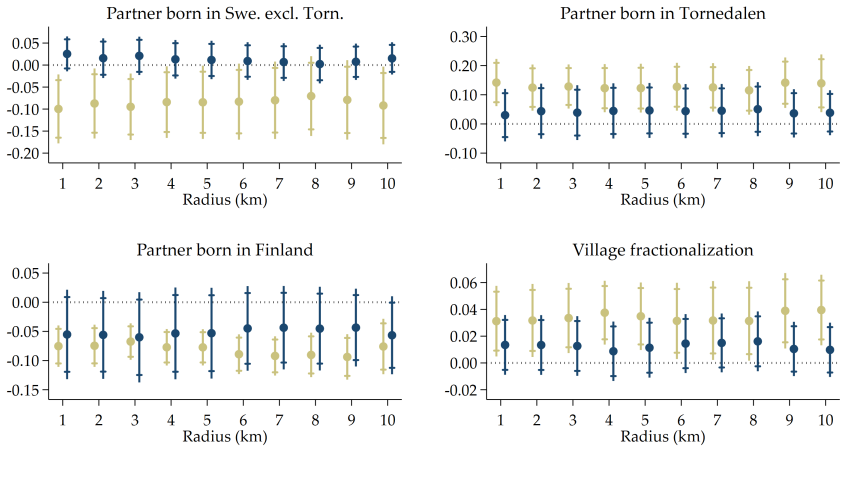
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Village-Level Birthplace Shares

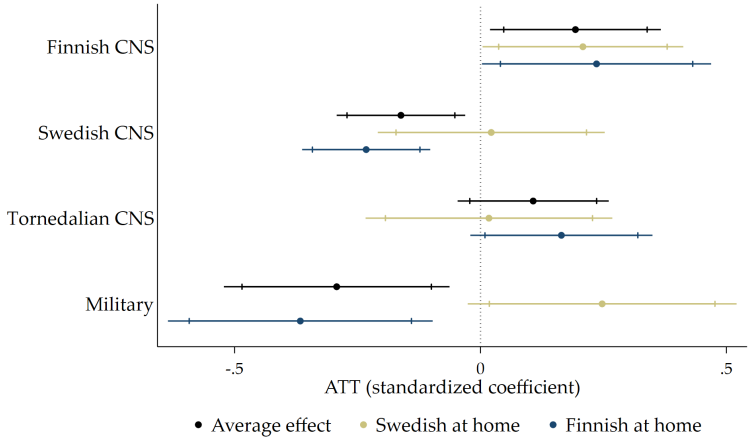
	Share born in		
	Tornedalen	Finland	Swe. excl. Torn.
	(1)	(2)	(3)
Treated × Post	-0.029	0.006	0.019
	(0.022)	(0.007)	(0.022)
Dep. var. mean	0.938	0.029	0.031
Observations	577	577	577
Villages	150	150	150
Village FE	✓	✓	✓
Year FE	✓	✓	✓

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Social Integration – Heterogeneity Share Recorded as “Finnish”



Heterogeneity by language use at home



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